

Ram Singh
Mob: 9463067522
E-mail: bhankharz@csepup.ac.in

CPE-202 Python Programming
MST-I/11-2020

Navdeep Singh
Mob: 7696353152
E-mail: navdeepsony@csepup.ac.in

Time Allowed: 60 Minutes

Max. Marks: 15

Section-A is compulsory. Attempt any two from section-B

SECTION-A

Q1:

1x5=5

- An object O=[3,4,4,5,5,5,6,6,7] is given. Write python statement to store it in a variable V which may contain only distinct data elements.
- An object A hold data element as [2, 3, 4, 6, 8, 10]. Write python object comprehension statement to make these values in A as [5, 8, 11, 17, 23, 29].
- An 3x4 numeric array contains data elements as [[1,1,1,1],[2,2,2,2],[3,3,3,3]]. Write Python method to store this array as 6x2 and then extract an sub-array of [[2,2],[3,3]] from the 6x2 array.
- What will be output of Python expression x={i: i**2 for i in range(1,10) if i%2!=0} ?
- Giving example of each, describe how list(), tuple(), set() and dict() objects are created ?

SECTION-B

Q2 Implement the following equation in Python:

$$S = \sum_{i=1}^N V_{(i\%2==0)*3}$$

where i=1 to 20

Q2 A company stores its employee's data in the following structure:

EMPID	EMPNAME	EMPDESIG	DEPTNO	MOBNO
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Create a dictionary to implement the same with data of your choice.

Q3 Create two matrix A and B of 3x3 size using list(). Write possible Python procedure of your choice to compute product of both the matrix.

**Department of Computer Sci. & Engg.,
Punjabi University, Patiala
First Mid-Semester Test (Oct 2022)**

B.Tech-II (CSE) CPE-202
Time: 1 hour

Sem: 3rd
Python Programming

MM: 15

Section - A (1*5=5)

12101078 Prem Kumar

1. Write briefly various features of Python.
2. Differentiate between List and a tuple with the help of an example.
3. What are shared references? Explain with the help of an example.
4. For the dictionary $\hat{D}=\{'a':1,'b':2,'c':[10,20,30], 'd':{'e':3,'f':4}\}$, write the syntax for displaying the value 20 belonging to the list 'c' and value of 'f' belonging to the dictionary 'd'.
5. Differentiate between global and local variables with the help of an example.

Section - B (5*2=10) (Do any two)

1. For the following numpy arrays A and B:

A = np.array([1,2,3,4,5,6,7])

B = np.array([[1,2,3,4,5],[6,7,8,9,10]])

Write python statements to:

- i) Slice elements from index 3 to index 1 in A
- ii) Slice elements 6 and 7 from A
- iii) Slice elements [[2,3],[7,8]] from B
- iv) Print the last element 10 from B
- v) Return the first rows of the last two 2-dim array from the following 3-dim array:

```
array([[[ 0,  1,  2,  3,  4],  
       [ 5,  6,  7,  8,  9],  
       [10, 11, 12, 13, 14]],
```

```
      [[15, 16, 17, 18, 19],  
       [20, 21, 22, 23, 24],  
       [25, 26, 27, 28, 29]],
```

```
      [[30, 31, 32, 33, 34],  
       [35, 36, 37, 38, 39],  
       [40, 41, 42, 43, 44]]])
```

2. What is a class? Write a program to create a class that finds the sum of two numbers. The class should have
 - a. a constructor that takes two arguments used to initialize the variables 'a' and 'b'.
 - b. a member function named 'display' used to display the sum of two numbers.
3. Write a Python program that accepts a sentence and calculates the number of uppercase letters and lowercase letters.

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MST-I
CSEB2301T: Python Programming
Department of Computer Science & Engineering

Time Allowed: 60 Minutes

Note: Section A is compulsory. Attempt *any two* questions from Section-B.

Max. Marks: 15

SECTION-A (5 * 1 = 5)

i) Differentiate between mutable and immutable objects with the help of an example.

ii)

```
def modify_list(lst):  
    lst[0] = lst[0] + 10  
    lst = [4, 5, 6]  
numbers = [1, 2, 3]  
modify_list(numbers)  
print(numbers)
```

iii).

```
class Counter:  
    count = 0  
    def __init__(self):  
        self.count = 1  
        Counter.count += 1  
obj1 = Counter()  
obj2 = Counter()  
print(obj1.count, obj2.count, Counter.count)
```

iv) Given a 2D NumPy array of shape (2,4), transform it into a 1D array of shape (1,8) without using loops.

v) Explain different types of inheritance in Python.

SECTION-B (2 * 5 = 10)

- vi) Write a program in which a function takes a string and returns a dictionary that counts the frequency of each character in the string.
- vii) Give examples of using these list methods: append, extend, pop, remove and sort.
- viii) Create a numpy matrix of shape (4,4) and find its transpose, inverse, determinant, diagonal and trace.

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Time Allowed: 60 Minutes

Note: Section A is compulsory. Attempt *any two* questions from Section-B.

SECTION-A (5 * 1 = 5)

What will be the output of these programs (2 Marks)

i). Differentiate between mutable and immutable objects with the help of an example. (1Mark)

ii). `def update_list(my_list):
 my_list.append(100)
 my_list = [1, 2, 3]
 my_list.append(200)`

`lst = [10, 20]
update_list(lst)
print(lst)`

b) `class Test:
 value = 5

 def __init__(self):
 self.value = Test.value
 Test.value += 1

t1 = Test()
t2 = Test()
print(t1.value, t2.value, Test.value)`

iii). Explain the difference between indexing and slicing. (1Mark)

iv). Differentiate between local and global variables with an example. (1Mark)

SECTION-B (2 * 5 = 10)

Q2. Write a program in which a function takes a string and returns a dictionary that counts the frequency of each character in the string.

Q3. Discuss various list operations with examples.

Q4. a) Explain the concept of list comprehension with an example.

b) Using a lambda function with the filter() method, filter out all the negative numbers from the list [-5, 3, -1, 7, -2, 0].

Q5 Explain the concept of Classes and Inheritance in Python with suitable examples.

CPE-202 Python Programming

MST-1

Time Allowed: 60 Minutes

MM: 15

Section-A is compulsory. Attempt any two questions from Section-B

Section-A

1. How can you concatenate two tuples? Write Python code for the same.
2. What is the output of `print list1 + list2`, if `list1 = [ 'akash', 985 , 2.23, 'deep' ]` and `list2 = [54, 'singh']`?
3. Write a program to test whether a given number is within user defined range or not?
4. Where do we need Python dictionaries?
5. Slice an array `[45,23,78,34; 35,12,9,6; 67,4,98,12; 43,67,34,2]` with slicing range from second element of second row to fourth element of third row.

Section-B

6.
 - a. Define dictionary. Define two sample dictionaries and merge them by using Python Code.
 - b. Differentiate between Tuple and List datatype. Write a Python program to create a list of tuples from given list having number and its cube in each tuple
7.
 - a. Write a Python program to find duplicates in an Array.
 - b. Python program for removing i-th character from a string
8. Create two matrix A and B of 3 x 3 size using `list()`. Write possible Python procedure of your choice to compute product of both the matrix.